Experiment 1: Smart Login Page with Attempt Tracker (Closures + Validation)

```
<!DOCTYPE html>
<html>
<head>
    <title>Simple Login</title>
    <style>
        body {
            font-family: Arial;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            background: rgb(241, 3, 43);
        }
        .box {
            background: rgb(255, 230, 5);
            padding: 20px;
            width: 300px;
            text-align: center;
        }
        input {
            width: 100%;
            padding: 8px;
```

```
            margin: 8px 0;
        }
        button {
            width: 100%;
            padding: 8px;
        }
        .error {
            color: red;
            font-size: 14px;
        }
        @media (max-width: 400px) {
            .box {
                width: 90%;
            }
        }
    </style>
</head>
<body>
<div class="box">
    <h3>Login</h3>
    <input type="text" id="user" placeholder="Username">
    <input type="password" id="pass" placeholder="Password">
    <div class="error" id="msg"></div>
    <button onclick="login()">Login</button>
```

```
</div>
<script>
    const button=document.querySelector(".btn");
    let maxAttempt=3;
    let attempts=0;
    button.addEventListener("click,()=>" {
        attempts++;
        if(attempts>=maxAttempt){
            button.disabled=true;
            alert("maximum attempts are reached");
        }
    })
    </script>
    </body>
    </html>
```

OUTPUT:

Experiment 2: Advanced Calculator using Callback Functions

```
<!DOCTYPE html>
<html>
<head>
<title>Simple Calculator</title>
<style>
body {
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
    font-family:Arial;
}
.calculator {
    padding:15px;
    background:#fff;
    box-shadow:0 0 10px #aaa;
    border-radius:10px;
}
input {
    width:100%;
    height:40px;
    margin-bottom:10px;
    text-align:right;
```

```
        font-size:18px;
    }
    button {
        width:50px;
        height:40px;
        margin:5px;
        font-size:16px;
        }
        .result {
            margin-top:10px;
            font-weight:bold;
            color:green;
        }
        .error {
            color: red;
        }
    </style>
    </head>
    <body>
    <div class="calculator">
    <input type="text" id="d" disabled>
    <br>
    <button onclick="d.value='">C</button>
    <button onclick="d.value+='/'">÷</button>
```

```
<button onclick="d.value+='*'">×</button>
<button onclick="d.value+='-'">−</button><br>
<button onclick="d.value+='7'">7</button>
<button onclick="d.value+='8'">8</button>
<button onclick="d.value+='9'">9</button>
<button onclick="d.value+='+'">+</button><br>
<button onclick="d.value+='4'">4</button>
<button onclick="d.value+='5'">5</button>
<button onclick="d.value+='6'">6</button>
<button onclick="d.value=eval(d.value)">=</button><br>
<button onclick="d.value+='1'">1</button>
<button onclick="d.value+='2'">2</button>
<button onclick="d.value+='3'">3</button><br>
<button onclick="d.value+='0'">0</button>
<button onclick="d.value+='.'">.</button>
</div>
<script>
let d = document.getElementById("d");
</script>
</body>
</html>
```

OUTPUT:

Experiment 3: Number Analyzer Dashboard

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Analyzer</title>
</head>
    <style>
        *{
            padding: 0%;
            margin: 0%;
        }
```

```
body{
    background-color: rgb(216, 19, 19);
    color: cornsilk;
}
.loginpage{
    padding: 15%;
    padding-left: 35%;
}
.input{
    border: 3px solid cornsilk;
    border-radius: 10%;
    display: block;
    width: 250px;
    height: 200px;
    padding: 5%;
    box-shadow: 5px 10px #fff;
}
.number{
    display: block;
    margin-left: 35px;
    margin-bottom: 5px;
    border-radius: 5%;
}
.btn{
```

```
            margin-top: 15%;
            margin-left: 35%;
            width: 75px;
            height: 30px;
            border-radius: 15%;
            border: 1px solid cornsilk;
        }
    </style>
<body>
    <div class="loginpage">
        <form action="#" class="input">
            <label class="number" for="number">Enter a
Number</label>
            <input type="number" id="number" name="number"
class="number">
            <button class="btn">submit</button>
            <p class="analyzer"></p>
        </form>
    </div>
<script>
document.querySelector(".input").addEventListener("submit",
function(event) {
    event.preventDefault();
```

```
    let num =
parseInt(document.getElementById("number").value);

    let result = document.querySelector(".analyzer");


    if (isNaN(num)) {

        result.innerHTML = "Please enter a valid number";

        return;

    }

    let palindrome = isPalindrome(num) ? "Palindrome" : "Not
Palindrome";

    let armstrong = isArmstrong(num) ? "Armstrong" : "Not
Armstrong";

    let prime = isPrime(num) ? "Prime" : "Not Prime";

    let fact = factorial(num);

    result.innerHTML =

        "Palindrome: " + palindrome + "<br>" +

        "Armstrong: " + armstrong + "<br>" +

        "Prime: " + prime + "<br>" +

        "Factorial: " + fact;

});

function isPalindrome(num) {

    let str = num.toString();

    let reversed = str.split("").reverse().join("");

    return str === reversed;

}
```

```
function isArmstrong(num) {
    let str = num.toString();
    let power = str.length;
    let sum = 0;
    for (let i = 0; i < str.length; i++) {
        sum += Math.pow(parseInt(str[i]), power);
    }
    return sum === num;
}
function isPrime(num) {
    if (num <= 1) return false;
    for (let i = 2; i <= Math.sqrt(num); i++) {
        if (num % i === 0) return false;
    }
    return true;
}
function factorial(num) {
    if (num < 0) return "Not defined for negative numbers";
    let fact = 1;
    for (let i = 1; i <= num; i++) {
        fact *= i;
    }
```

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1979

```
    return fact;
}
</script>
</body>
</html>
```

OUTPUT:

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